

Hospital Resource Management USING CRUD operations.

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Abstract

The Hospital Resource Management System is designed to efficiently manage hospital resources such as patients, doctors, staff, rooms, and medical equipment using CRUD (Create, Read, Update, Delete) operations. The system aims to digitize traditional manual processes, reducing errors, improving data accessibility, and enhancing operational efficiency. This project provides a user-friendly interface to store, retrieve, update, and delete

records, ensuring better resource allocation and management within a hospital environment.

Introduction

Hospitals handle a large amount of data daily, including patient records, doctor schedules, and equipment

usage. Managing this information manually can lead to inefficiencies and errors. A Hospital Resource Management System automates these processes using CRUD operations, enabling structured data handling.

CRUD operations form the backbone of this system:

- **Create** – Add new records
- **Read** – View existing records
- **Update** – Modify records
- **Delete** – Remove records

This system improves accuracy, saves time, and enhances decision-making in healthcare institutions.

Objectives

- To design a system that efficiently manages hospital resources.
 - To implement CRUD operations for handling data.
 - To reduce manual paperwork and human errors.
 - To provide quick access to patient and resource information.
 - To ensure data consistency and security.
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Scope of the Project

The system focuses on managing:

- Patient details (admission, discharge, medical records)
- Doctor information (specialization, availability)
- Staff records
- Room allocation
- Equipment management

It is suitable for small to medium-sized hospitals and can be expanded for larger institutions.

Literature Review

Existing hospital systems often rely on manual records or partially digital systems, leading to inefficiencies. Studies show that digital healthcare management systems improve operational

efficiency and reduce errors. CRUD-based applications are widely used due to their simplicity and effectiveness in database management. This project builds on these concepts to provide a structured and scalable solution.

Methodology

The project follows a structured development approach:

1. **Requirement Analysis** – Understanding hospital needs.
 2. **System Design** – Creating database and UI structure.
 3. **Implementation** – Coding CRUD operations.
 4. **Testing** – Verifying system functionality.
 5. **Deployment** – Making the system available for use.
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Technology Used

- **Programming Language:** C / C++ / Java / Python (based on implementation)
- **Database:** MySQL / SQLite
- **Frontend:** HTML, CSS (if web-based)
- **Backend:** PHP / Python / Java
- **Tools:** VS Code / CodeBlocks / Eclipse

Architecture / System Design

The system follows a **three-tier architecture**:

1. **Presentation Layer** – User interface for interaction.
2. **Application Layer** – Business logic handling CRUD operations.
3. **Database Layer** – Stores hospital data.

Modules:

- Patient Management
- Doctor Management
- Staff Management
- Resource Allocation
- Reporting System

Implementation Details

- **Create:** Add new patient, doctor, or resource records.
- **Read:** Display stored data in tabular format.
- **Update:** Modify existing records such as patient details or room allocation.
- **Delete:** Remove records when no longer needed.

Example functionalities:

- Patient registration
- Doctor assignment

- Room booking
- Equipment tracking

Results / Output

- Efficient data storage and retrieval
- Reduced manual errors
- Faster access to hospital information
- Improved resource utilization

Sample Output:

- Patient list displayed with details
- Doctor availability status
- Room allocation chart

Conclusion

The Hospital Resource Management System using CRUD operations successfully simplifies hospital data management. It enhances efficiency, reduces workload, and ensures accurate record-keeping. The system demonstrates how basic database operations can be used to build practical real-world applications.

Future Work

- Integration with mobile applications

- Cloud-based data storage
 - Advanced analytics and reporting
 - AI-based patient prediction systems
 - Multi-hospital connectivity
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Footnotes / References

1. Database Management System Concepts – Silberschatz, Korth
 2. Software Engineering – Pressman
 3. Online tutorials and documentation (W3Schools, GeeksforGeeks)
 4. Research papers on healthcare management systems
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